What is Quantum in Grover's and Shor's Algorithms?
(Session 5 – Day 2, 4 July 2026)

Dr. Shashank Gupta

Learning Objectives

- Understand the principles of amplitude amplification and phase reflection in Grover's search algorithm.
- Master the geometric analysis of Grover's search as a 2D rotation of state vectors.
- Comprehend Shor's factoring algorithm, specifically the reduction of factoring to period-finding.
- Learn the math behind the Quantum Fourier Transform (QFT) and its circuit implementation.
- Compare the query and time complexities of classical vs. quantum search and factoring.

1 Grover's Search Algorithm

Grover's algorithm solves the problem of searching an unstructured database of size $N = 2^n$ to locate a unique target state $|\omega\rangle$ that satisfies a black-box function $f(\omega) = 1$ and $f(x) = 0$ for all $x \neq \omega$.

1.1 Complexity Comparison

* **Classical search:** Requires $\mathcal{O}(N)$ queries in the worst case (and $N/2$ on average). * **Quantum search:** Requires $\mathcal{O}(\sqrt{N})$ queries.

1.2 The Grover Iteration

We start with the uniform superposition state:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

A single Grover iteration consists of two operations:

1. **Phase Oracle (U_ω):** Flips the sign of the target state:
$$U_\omega = I - 2\,|\omega\rangle\langle\omega| \implies U_\omega\,|x\rangle = (-1)^{f(x)}\,|x\rangle$$
2. **Grover Diffusion Operator (D):** Reflects the state vector about the uniform superposition state $|s\rangle$:
$$D = 2\,|s\rangle\langle s| - I = H^{\otimes n}(2\,|0^n\rangle\langle 0^n| - I)H^{\otimes n}$$

1.3 Geometric Analysis in 2D Subspace

We can analyze Grover's search in the 2D subspace spanned by the target state $|\omega\rangle$ and the normalized superposition of all non-target states $|s'\rangle$:

$$|s'\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq \omega} |x\rangle$$

The initial uniform superposition state $|s\rangle$ can be written as:

$$|s\rangle = \sin\left(\frac{\theta}{2}\right)|\omega\rangle + \cos\left(\frac{\theta}{2}\right)|s'\rangle$$

where $\sin(\theta/2) = \frac{1}{\sqrt{N}}$. Each application of the Grover iteration $G = DU_\omega$ rotates the state vector in this 2D plane towards the target state $|\omega\rangle$ by an angle θ:

$$G^k|s\rangle = \sin\left(\frac{2k+1}{2}\theta\right)|\omega\rangle + \cos\left(\frac{2k+1}{2}\theta\right)|s'\rangle$$

To maximize the probability of measuring the target state, we choose k such that $\frac{2k+1}{2}\theta \approx \frac{\pi}{2}$. This yields the optimal number of iterations:

$$R \approx \frac{\pi}{4}\sqrt{N}$$

2 Shor's Factoring Algorithm

Shor's algorithm factors a composite integer N in polynomial time $\mathcal{O}((\log N)^3)$. Factoring is reduced to finding the period of a modular function, which can be done efficiently on a quantum computer.

2.1 Classical Reduction of Factoring to Period-Finding

To factor N:

1. Choose a random integer $a < N$ and compute $\gcd(a, N)$. If $\gcd(a, N) > 1$, we have found a factor.

2. Otherwise, define the modular function $f(x) = a^x \pmod N$. Find its period r, which is the smallest integer $r > 0$ such that:
$$a^r \equiv 1 \pmod N$$

3. If the period r is even and $a^{r/2} \not\equiv -1 \pmod N$, then the non-trivial factors of N are:

$$p = \gcd(a^{r/2} - 1, N), \qquad q = \gcd(a^{r/2} + 1, N)$$

2.2 Quantum Period-Finding via Phase Estimation

To find the period r of $U|y\rangle = |ay \pmod{N}\rangle$:

1. We initialize two registers: the first (estimation) register with t qubits in state $|0\rangle^{\otimes t}$, and the second (target) register in state $|1\rangle$.
2. We prepare a superposition in the first register and apply controlled-power unitary operators:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^t}} \sum_{x=0}^{2^t-1} |x\rangle |a^x \pmod{N}\rangle$$

3. Applying the Inverse Quantum Fourier Transform ($\text{QFT}^\dagger$) to the first register rotates the state, projecting out the phase $\phi \approx k/r$.
4. Measuring the first register yields the phase, and we use the continued fractions algorithm to reconstruct r with high probability.

2.3 The Quantum Fourier Transform

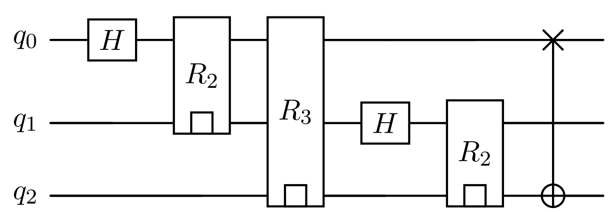
The QFT acts on a basis state $|j\rangle$ of n qubits (dimension $M = 2^n$) as:

$$\text{QFT}|j\rangle = \frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{2\pi i jk/M} |k\rangle$$

The circuit implementation uses Hadamard gates and controlled-phase rotation gates R_k:

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$$

The circuit diagram for the 3-qubit QFT is:



3 Discussion Problems / Exercises

1. **Grover Diffuser Unitary Proof:** Prove that the Grover diffuser $D = 2|s\rangle\langle s| - I$ is unitary ($D^\dagger D = I$) and that it acts as a reflection.

> **Session Takeaway:** Because $\langle s|s\rangle = 1$, squaring the reflection operator yields $D^2 = I$. Since D is also self-adjoint ($D^\dagger = D$), we have $D^\dagger D = I$, proving it is a valid quantum operator.

2. **Shor's Factoring for $N = 15$:** For $N = 15$ and chosen base $a = 7$, write down the powers $7^x \pmod{15}$ to determine the period r, and show how the factors are retrieved.

   > **Session Takeaway:** The period of $7^x \pmod{15}$ is $r = 4$. Since r is even and $7^{4/2} = 49 \equiv 4 \not\equiv -1 \pmod{15}$, we compute $\gcd(4-1, 15) = 3$ and $\gcd(4+1, 15) = 5$ to retrieve the factors.

3. **QFT Matrix for $n = 2$:** Construct the explicit 4×4 unitary matrix of the QFT for $n = 2$ qubits.

   > **Session Takeaway:** The QFT matrix for $n = 2$ corresponds to the discrete Fourier transform matrix F_4 with elements $F_{jk} = \frac{1}{2} i^{jk}$ for $j, k \in \{0, 1, 2, 3\}$.

4 Take-away Messages

1. Grover's algorithm uses phase reflection and amplitude amplification to locate a target state with quadratic query speed-up.
2. Shor's algorithm achieves exponential speed-up by translating arithmetic factoring to period-finding, solved using the QFT.
3. The QFT does not output all classical Fourier coefficients; instead, it generates a state whose amplitudes correspond to the Fourier transform, allowing us to sample global periodic properties.

Further reading (optional)
Nielsen & Chuang, Chap. 5 & 6;
Qiskit Textbook, "Shor's Algorithm";
PennyLane documentation, "Grover's Algorithm".

A Detailed Solutions to Discussion Problems

A.1 Solution to Problem 1: Unitarity of the Grover Diffuser

We define the operator $D = 2\ket{s}\bra{s} - I$. Since the state $\ket{s}$ is normalized, we have $\braket{s|s} = 1$. 1. **Self-adjoint check:**

$$D^\dagger = (2\ket{s}\bra{s} - I)^\dagger = 2(\bra{s})^\dagger(\ket{s})^\dagger - I^\dagger = 2\ket{s}\bra{s} - I = D$$

Thus, D is Hermitian. 2. **Unitarity check:**

$$D^\dagger D = D^2 = (2\ket{s}\bra{s} - I)(2\ket{s}\bra{s} - I)$$

Multiplying out the terms:

$$D^2 = 4\ket{s}\braket{s|s}\bra{s} - 2\ket{s}\bra{s} - 2\ket{s}\bra{s} + I$$

Since $\braket{s|s} = 1$:

$$D^2 = 4\ket{s}\bra{s} - 4\ket{s}\bra{s} + I = I$$

Since $D^\dagger D = I$, the operator is unitary. Geometrically, for any state vector $\ket{\psi} = a\ket{s} + b\ket{s_\perp}$, we have:

$$D\ket{\psi} = D(a\ket{s} + b\ket{s_\perp}) = a(2\ket{s}\braket{s|s} - \ket{s}) + b(2\ket{s}\braket{s|s_\perp} - \ket{s_\perp}) = a\ket{s} - b\ket{s_\perp}$$

This flips the component perpendicular to $\ket{s}$, which is a reflection about the state vector $\ket{s}$.

A.2 Solution to Problem 2: Shor's Factoring for $N = 15$, $a = 7$

We want to find the period r of $f(x) = 7^x \pmod{15}$. 1. We compute $7^x \pmod{15}$ sequentially: * $x = 0: \quad 7^0 \pmod{15} = 1$ * $x = 1: \quad 7^1 \pmod{15} = 7$ * $x = 2: \quad 7^2 \pmod{15} = 49 \pmod{15} = 4$ * $x = 3: \quad 7^3 \pmod{15} = 4 \times 7 \pmod{15} = 28 \pmod{15} = 13$ * $x = 4: \quad 7^4 \pmod{15} = 13 \times 7 \pmod{15} = 91 \pmod{15} = 1$ Since $7^4 \equiv 1 \pmod{15}$, the sequence repeats. The period is $r = 4$. 2. We check the factorization conditions: * Is r even? Yes, $r = 4$. * Is $7^{r/2} \equiv 7^2 = 49 \equiv 4 \not\equiv -1 \pmod{15}$? Yes, $4 \not\equiv 14 \pmod{15}$. 3. We compute the factors: * $p = \gcd(7^2 - 1, 15) = \gcd(48, 15)$ Applying Euclid's algorithm: $\gcd(48, 15) = \gcd(3, 15) = 3$. * $q = \gcd(7^2 + 1, 15) = \gcd(50, 15)$ Applying Euclid's algorithm: $\gcd(50, 15) = \gcd(5, 15) = 5$. We have successfully found the factors $15 = 3 \times 5$.

A.3 Solution to Problem 3: Explicit QFT Matrix for $n = 2$

For $n = 2$, we have $M = 2^2 = 4$ states: $\ket{0}, \ket{1}, \ket{2}, \ket{3}$. The matrix elements are given by:

$$\text{QFT}_{jk} = \frac{1}{\sqrt{4}} e^{2\pi i jk/4} = \frac{1}{2} e^{i\pi jk/2} = \frac{1}{2} i^{jk}$$

Let's compute the elements for $j, k \in \{0, 1, 2, 3\}$: * **Row 0 ($j = 0$):** $i^{0\cdot k} = 1$ for all k. * **Row 1 ($j = 1$):** $i^{1\cdot k} = [1, i, -1, -i]$. * **Row 2 ($j = 2$):** $i^{2\cdot k} = [1, -1, 1, -1]$ (since $i^2 = -1$, $i^4 = 1$, $i^6 = -1$). * **Row 3 ($j = 3$):** $i^{3\cdot k} = [1, -i, -1, i]$ (since $i^3 = -i$, $i^6 = -1$, $i^9 = i$). Assembling the 4×4 matrix:

$$\text{QFT} = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

This is exactly the F_4 discrete Fourier transform matrix, which is unitary.